Soru Classify these system into linear and non linear

a) $y(t) = 2x(t) - 3$

b) $y(t) = e^{-t} x(t)$

Linearity cond : $x_1(t) \xrightarrow{T[\cdot]} y_1(t)$, $x_2(t) \xrightarrow{T[\cdot]} y_2(t)$ $\Big) T[ax_1(t) + bx_2(t)] \stackrel{?}{=} a\underbrace{T[x_1(t)]}_{y_1(t)} + b\underbrace{T[x_2(t)]}_{y_2(t)}$

a) $x_1(t) \longrightarrow y_1(t) = 2x_1(t) - 3$

$x_2(t) \longrightarrow y_2(t) = 2x_2(t) - 3$

$x_3(t) = ax_1(t) + bx_2(t)$

$\longrightarrow y_3(t) = 2x_3(t) - 3 = 2ax_1(t) + 2bx_2(t) - 3$

$\stackrel{?}{\neq} a(2x_1(t) - 3) + b(2x_2(t) - 3)$

b) $y(t) = e^{-t} x(t)$ non linear !

$x_1(t) \longrightarrow y_1(t) = e^{-t} x_1(t)$

$x_2(t) \longrightarrow y_2(t) = e^{-t} x_2(t)$

$x_3(t) = a x_1(t) + b x_2(t)$

$\longrightarrow y_3(t) = e^{-t} x_3(t) = e^{-t} ax_1(t) + e^{-t} bx_2(t)$

$\stackrel{?}{=} a\underbrace{y_1(t)}_{e^{-t}x_1(t)} + b\underbrace{y_2(t)}_{e^{-t}x_2(t)}$ ✓

linear system

Soru Determine whether the following systems are time variant or time invariant

a) $y(t) = 2x(t) + 3$

b) $y(t) = (t+2)x(t)$

Time invariant cond $x(t) \longrightarrow y(t)$

$x(t - t_0) \longrightarrow y(t - t_0)$

a) $x_1(t) \longrightarrow \overbrace{2x_1(t)+3}^{y_1(t)}$

$x_2(t) = x(t - t_0)$

$y_2(t) \stackrel{?}{=} y_1(t - t_0)$

$2x_2(t) + 3 = 2x(t - t_0) + 3 \stackrel{?}{=} 2x_1(t - t_0) + 3$

✓

zamanla değişmez

b) $x_1(t) \rightarrow y_1(t) = (t+2)\,x_1(t)$

$x_2(t) = x_1(t-t_0) \longrightarrow y_2(t) = (t+2)\,x_2(t)$

$y_2(t) = (t+2)\,x_1(t-t_0)$

$y_2(t) \overset{?}{=} y_1(t-t_0)$

$(t+2)\,x_1(t-t_0) \overset{?}{\neq} (t-t_0+2)\,x_1(t-t_0)$

time variant

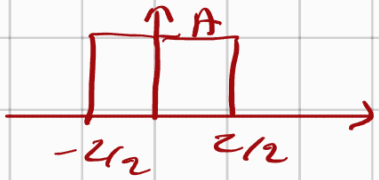
Soru: A system is causal if its input at any time t_0 depends on the input at times prior to t_0

$$y(t_0) = T[x(t): t \leq t_0]$$

A nessesary an sufficent condition for an LTI system to be causal is that its impulse response $h(t)$ must be causal signal for $t < 0$, we mast have $h(t) = 0$ (the output when the input is $\delta(t)$)

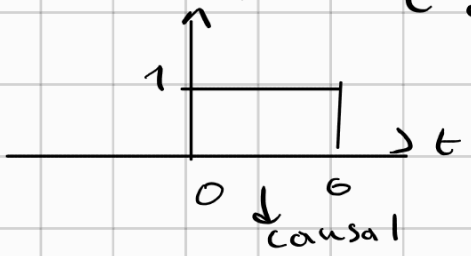
$h(t) = \Pi\left(\frac{t-3}{6}\right)$, LTI system is causal?

$\Pi(t/2) \rightarrow$





$h(t) = \Pi\left(\frac{t-3}{6}\right)$



Soru: Determine the Fourier Series Expansion of the following signals.

a) $x(t) = \sum_{n=-\infty}^{\infty} \Lambda(t-n)\, u(t-n)$

b) $x(t) = \sum_{n=-\infty}^{\infty} (-1)^n \, \delta(t-nT)$

Fourier Seri Açılımları

Trigonometrik FS: $x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t))$

$$a_0 = \frac{1}{T_0}\int_{T_0} x(t)\,dt, \quad a_n = \frac{2}{T_0}\int x(t)\cos(n\omega_0 t)\,dt$$

$$, \; b_n = \frac{2}{T_0}\int_{T_0} x(t)\sin(n\omega_0 t)\,dt$$

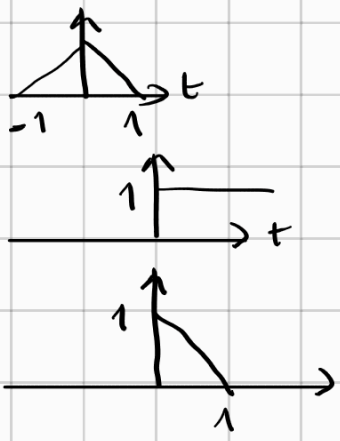
Kompleks ExpFS: $x(t) = \sum_{n=-\infty}^{\infty} x_n e^{jn\omega_0 t}$

$$x_n = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-jn\omega_0 t}\,dt$$

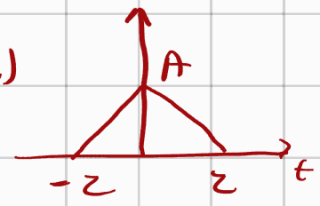
> ikisiarasındaki ilişki
> $x_n = \frac{a_n - jb_n}{2}$

a) önce n=0 ve =1 için çizilsin

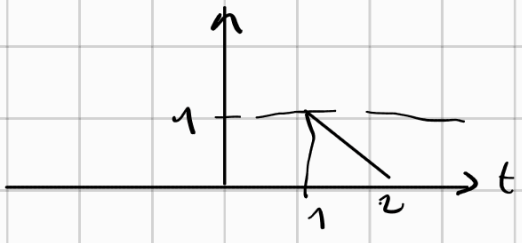
$n=0 \rightarrow \Lambda(t)\,u(t)$



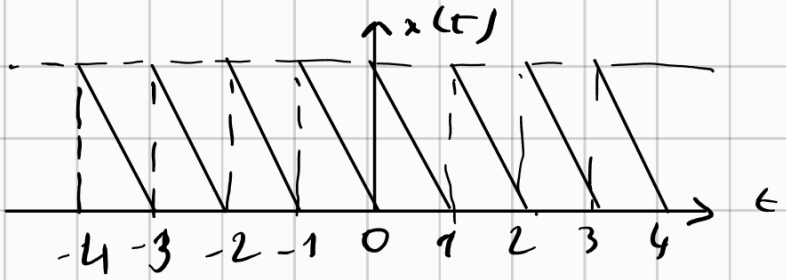
$A\Lambda(t/2)$



$n=1 \rightarrow \Lambda(t-1)\,u(t-1)$



$\sum_{n=-\infty}^{\infty} \Lambda(t-n)\,u(t-n)$



kompleks FS

$$x_0 = \frac{1}{T_0}\int_{T_0} x(t)\,dt$$

$$= \frac{1}{T_0}\int x(t)\,dt$$

$$= \int_0^1 (1-t)\,dt = \frac{1}{2}$$

$T_0 = 1$

$y - y_0 = m(x - x_0)$ (doğrunun eğimi)

$m = -1$

$y - 1 = (-1)x$

$y(t) = 1 - t$

$T_0 = 1$

$n \neq 0 \rightarrow x_n = \int_0^1 x(t)\, e^{-jn2\pi t} = \int_0^1 (1-t)\, e^{-jn2\pi t}\,dt = \int_0^1 e^{-jn2\pi t}\,dt - \underbrace{\int_0^1 t\, e^{-jn2\pi t}\,dt}_{I_1}$

$\omega_0 = \frac{2\pi}{T_0}$

$$-\frac{1}{jn2\pi}\, e^{-jn2\pi t}\Big|_0^1 = \frac{-1}{jn2\pi}\left[e^{-jn2\pi} - e^0\right]$$

$$I_1 = \int_0^1 t\, e^{-jn2\pi t}\,dt$$

kısmi integrasyon

$\int_a^b uv = uv\Big|_a^b - \int_a^b v\,du$

$u = t$ $\qquad e^{-jn2\pi t}\,dt = dv$

$du = dt$ $\qquad \frac{-1}{jn2\pi}\, e^{-jn2\pi t} = v$

$$= \left(t \cdot \frac{-1}{jn2\pi}\, e^{-jn2\pi t}\right)\Big|_0^1 - \int_0^1 -\frac{1}{jn2\pi}\, e^{-jn2\pi t}\,dt$$

$$= -\frac{1}{jn2\pi}\, e^{-jn2\pi} + \frac{1}{n^2 4\pi^2}\, e^{-jn2\pi} - \frac{1}{n^2 4\pi^2}$$

$$x_n = \frac{1}{jn2\pi} + \frac{1}{n^2 4\pi^2} - \frac{1}{n^2 4\pi^2}\, e^{-jn2\pi}$$

$$= \frac{1}{jn2\pi} + \frac{1}{n^2 4\pi^2}\left[1 - \underbrace{e^{-jn2\pi}}_{1}\right] = \frac{1}{jn2\pi}$$

Üstel FS: $\boxed{x(t) = \frac{1}{2} + \sum_{n\neq 0} \frac{1}{jn2\pi}\, e^{jn2\pi t}}$

$x_n = \frac{a_n - jb_n}{2}$ $\xrightarrow{-j \text{ ile çarpalım}}$ $\frac{1}{jn\pi} = -\frac{j}{n\pi}$

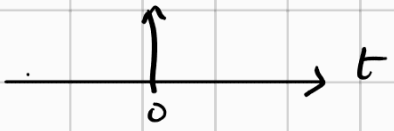
$a_n = 0$

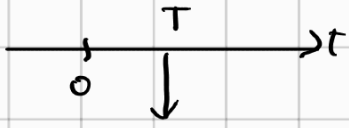
$b_n = \frac{1}{n\pi}$

$$x(t) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{1}{n\pi} \sin(\pi n t)$$

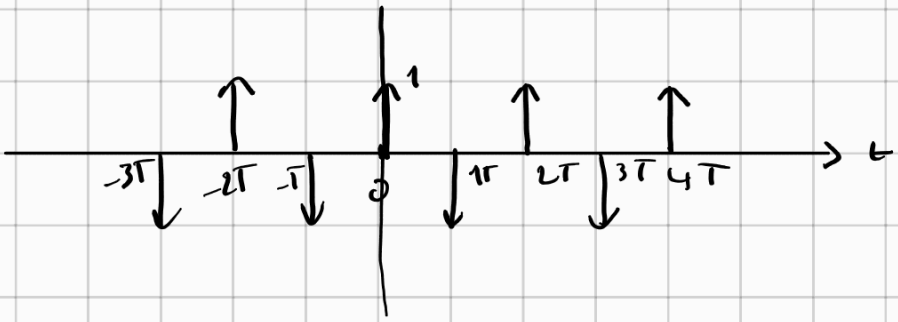
b) $n = 0 \rightarrow \delta(t)$

$n = 1 \rightarrow -\delta(t-T)$





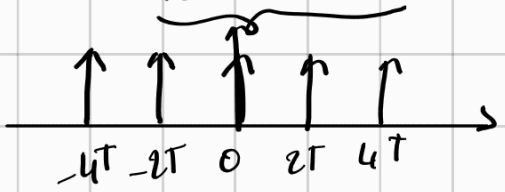
c)

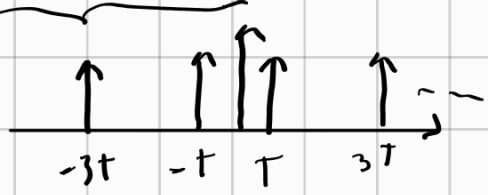


Dürtü katarının FS açılımı

$$x(t) = \sum_n \delta(t - nT_0) = \frac{1}{T_0} \sum_n e^{jn\omega_0 t}$$

$$x(t) = \sum_n \delta(t - n2T) - \sum_n \delta(t - T - n2T)$$





$$x(t) = \frac{1}{2T} \sum_n e^{jn\frac{\pi}{2T}t} - \frac{1}{2T} \sum_n e^{jn\frac{\pi}{2T}(t-T)}$$

$$= \frac{1}{2T} \sum_n \left\{ e^{jn\frac{\pi}{T}t} - e^{jn\frac{\pi}{T}t} \underbrace{e^{-jn\pi}}_{(-1)^n} \right\}$$

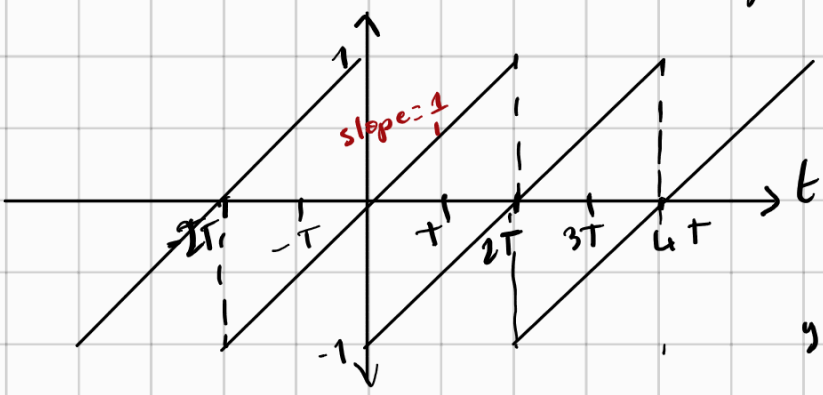
$e^{-jn\pi} = \cos(-n\pi) + j\sin(-n\pi) = (-1)^n$

$$= \frac{1}{2T} \sum_n e^{jn\frac{\pi}{T}\cdot t} \underbrace{(1 - (-1)^n)}$$

$$x_n = \frac{1}{2T}(1 - (-1)^n) = \begin{cases} \frac{1}{T} & n \text{ tek} \\ 0 & n \text{ çift} \end{cases}$$

$$\Rightarrow x(t) = \sum_n x_n e^{jn\frac{\pi}{T}\cdot t}, \quad x_n = \begin{cases} 1/T, & n \text{ tek} \\ 0, & n \text{ çift} \end{cases}$$

Soru: Determine the FS expansion of the following periodic signal



$$y = \frac{1}{T}x$$

$$x(t) = \frac{t}{T}$$

$$x_n = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-jn\omega_0 t}\, dt$$

$$T_0 = 2T \rightarrow \omega_0 = \frac{2\pi}{2T} = \frac{\pi}{T}$$

$$x_n = \frac{1}{2T}\int_{-T}^{T} \frac{t}{T} e^{-jn\frac{\pi}{T}} dt$$

$$= \frac{1}{2T^2}\int_{-T}^{T} e^{-jn\frac{\pi t}{T}} dt$$

$u = t$ $\quad e^{-jn\frac{\pi t}{T}} dt = dv$

$du = dt$ $\quad v = e^{-jn\frac{\pi t}{T}}$

$$= \frac{1}{2T^2}\left[\left(t \cdot \frac{1}{-jn\frac{\pi}{T}} e^{-jn\frac{\pi t}{T}} \right|_{-T}^{T} \right) - \int_{-T}^{T} -\frac{1}{jn\frac{\pi}{T}} e^{-jn\frac{\pi t}{T}} dt \Bigg]$$

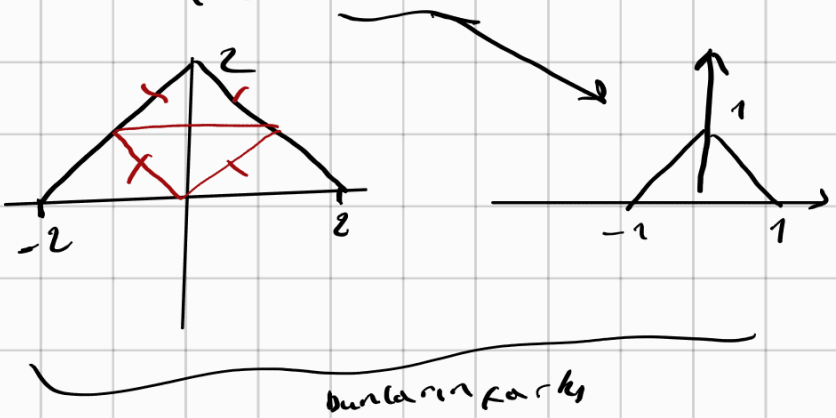
Sonuca yaklaşırken $e^{jn\pi} = (-1)^n$

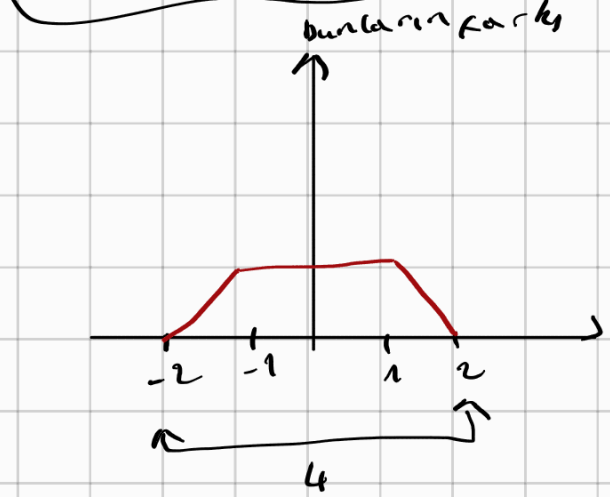
x_0'ı ayrı olarak bul ödev

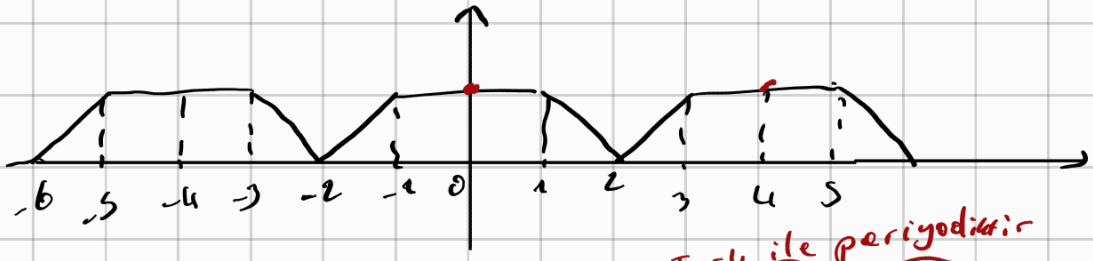
Soru Classify the following signal into periodic or non periodic

$$x(t) = 2 \sum_n \Lambda\left(\frac{t-4n}{2}\right) - \sum_n \Lambda(t-4n)$$

$n=0 \rightarrow x_1(t) = x_2(t) = 2\Lambda\left(\frac{t}{2}\right) - \Lambda(t)$

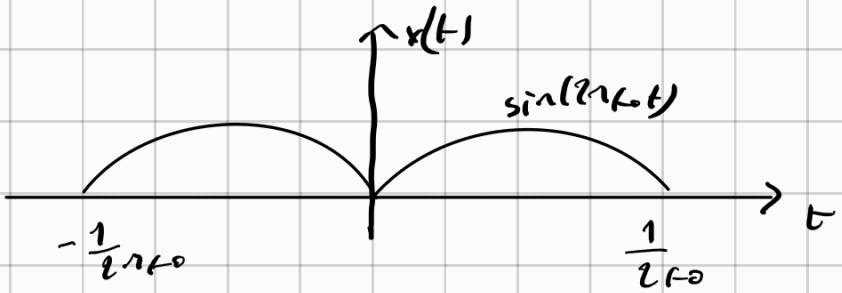




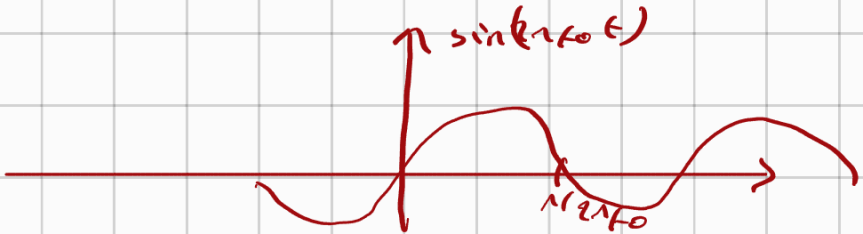


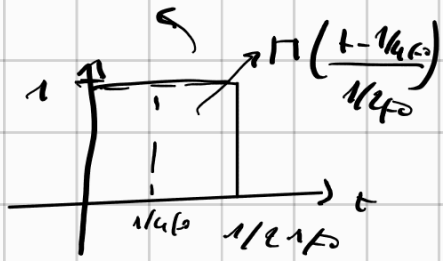
$T_0 = 4$ ile periyodiktir

Soru: Find the fourier Transform of the following signal



$X(f) = ?$





$$x(t) = \sin(2\pi f_0 t)\cdot \Pi\left(\frac{t-1/4f_0}{1/2f_0}\right) - \sin(2\pi f_0 t)\,\Pi\left(\frac{t+1/4f_0}{1/2f_0}\right)$$

$$x(t) = \underbrace{\sin(2\pi f_0 t)}_{x_1(t)}\left[\underbrace{\Pi\left(2f_0\left(t-\tfrac{1}{4f_0}\right)\right) - \Pi\left(t+\tfrac{1}{4f_0}\right)}_{x_2(t)}\right]$$

$$x(t) = x_1(t)\cdot x_2(t)$$

$$\downarrow$$

$$X(f) = X_1(f) \circledast X_2(f)$$

$$\sin(2\pi f_0 t) \xrightarrow{\mathcal{F}} \frac{-1}{2j}\left[\delta(f+f_0) - \delta(f-f_0)\right] \leftarrow X_1(f)$$

$$A\,\Pi(t/\tau) \xrightarrow{\mathcal{F}} A\tau\,\mathrm{sinc}(f\tau)$$

$$x(t-t_0) \rightarrow X(f)\cdot e^{-j2\pi f t_0}$$

$$x_2(t) = \Pi\left(2f_0\left(t-\underbrace{\frac{1}{4f_0}}_{t_0=\frac{1}{4f_0}}\right)\right) - \Pi\left(2f_0\left(t+\underbrace{\frac{1}{4f_0}}_{t_0=-\frac{1}{4f_0}}\right)\right)$$

$$\tau = \frac{1}{2f_0}$$

$$X_2(f) = \frac{1}{2f_0}\sin\left(f\cdot\frac{1}{2f_0}\right)e^{-j2\pi f\cdot\frac{1}{4f_0}}$$

$$= \frac{1}{2f_0}\,\mathrm{sinc}\left(f\cdot\frac{1}{2f_0}\right)\cdot e^{j2\pi f\cdot\frac{1}{4f_0}}$$

$$= \frac{j}{f_0}\cdot\mathrm{sinc}\left(\frac{f}{2f_0}\right)\sin\left(\frac{\pi f}{2f_0}\right)$$

$$X(f) = X_2(f) \circledast \left[\frac{1}{2j}\left(\delta(f-f_0) - \delta(f+f_0)\right)\right]$$

$$= \frac{1}{2j}\underbrace{X_2(f)\circledast\delta(f-f_0)}_{X_2(f-f_0)} - \frac{1}{2j}\underbrace{X_2(f)\circledast\delta(f+f_0)}_{X_2(f+f_0)}$$

$$x(t)\,\delta(t-t_0) = x(t_0)\cdot\delta(t-t_0)$$

$$x(t)\circledast\delta(t-t_0) = x(t-t_0)$$

Soru $x(t) = 4\Lambda\left(\frac{t}{4}\right)\cos(2\pi f_0 t)$

Find the Fourier Transform

Reminder: $A\Pi(t/\tau) \rightarrow A\tau \sin f\tau$

$\cos(2\pi f_0 t) \rightarrow \frac{1}{2}\delta(f-f_0) + \frac{1}{2}\delta(f+f_0)$

$v(t)\cos(2\pi f_0 t) \rightarrow \frac{1}{2}[X(f-f_0) + X(f+f_0)]$

Soru: $x(t) = \Lambda(2t+3) - \Lambda(3t-2)$ Find the fourier transform

Reminder $A\Lambda(t/\tau) \rightarrow A\tau \operatorname{sinc}^2(f\tau)$

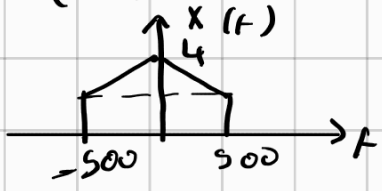
$x(t-t_0) \rightarrow X(f)e^{-j2\pi f t_0}$

$x(t) = \Lambda\left(\frac{t+3/2}{1/2}\right) - \Lambda\left(\frac{t-2/3}{1/3}\right)$

$A=1$, $\tau = 1/2$, $t_0 = -3/2$ | $A=1$, $\tau = 1/3$, $t_0 = 2/3$

$X(f) = \frac{1}{2}\operatorname{sinc}^2(f/2)e^{j2\pi f 3/2} - \frac{1}{3}\operatorname{sinc}^2(f/3)e^{-j2\pi f 2/3}$

Soru Fourier transform of a signal is



Determine and sketch the fourier transform of the signal

$x_1(t) = -x(t) + x(t)\cos(2000\pi t) + 2x(t)\cos^2(3000\pi t)$

$2\pi f_0 t$; $\frac{1}{2}[1+\cos(6000\pi t)]$

$$X_1(f) = \frac{1}{2}x(f-1000) + \frac{1}{2}x(f+1000) + \frac{1}{2}x(f-3000) + \frac{1}{2}x(f+3000)$$

$$\cos^2\alpha = \frac{1}{2}(1+\cos 2\alpha)$$

Soru: $x(t) = t\,\mathrm{sinc}(t)$

Find the fourier transform

1.yol

$$\mathrm{sinc}(t) = \frac{\sin(\pi t)}{\pi t} \longrightarrow 2\pi f_0 t$$

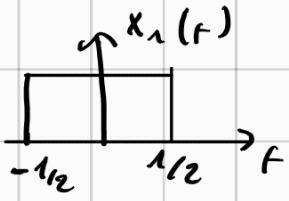
$$x(t) = \frac{1}{\pi}\sin(\pi t) \qquad 2\pi f_0 t = \pi t \qquad f_0 = ? = 1/2$$

$$X(f) = \frac{1}{2j\pi}\delta\left(f-\frac{1}{2}\right) - \frac{1}{2j\pi}\delta\left(f+\frac{1}{2}\right)$$

2.yol

Reminder: $t\,x(t) \longrightarrow \frac{j}{2\pi}\frac{dX(f)}{df}$

$$x_1(t) = \mathrm{sinc}(t) \longrightarrow X_1(f) = \Pi(f)$$



$$X(f) = u\left(f-\frac{1}{2}\right) - u\left(f+\frac{1}{2}\right)$$

$$\frac{dX_1(f)}{df} = \delta\left(f+\frac{1}{2}\right) - \delta\left(f-1/2\right)$$

$$X(f) = \frac{j}{2\pi}\left[\delta\left(f+\frac{1}{2}\right) - \delta\left(f-\frac{1}{2}\right)\right]$$

Soru: $x(t) = \mathrm{sinc}(t) \circledast \mathrm{sinc}(t) = \mathrm{sinc}(t)$

Prove.

$$F\{x_1(t) * x_2(t)\} = X_1(f)\cdot X_2(f)$$

$$x_1(t) = \mathrm{sinc}(t) \longrightarrow X_1(f) = \Pi(f)$$

$$x_2(t) = \mathrm{sinc}(t) \longrightarrow X_2(f) = \Pi(f)$$



$$X(f) = \Pi(f)$$

$$\downarrow \mathcal{F}^{-1}$$

$$x(t) = \mathrm{sinc}(t)$$

Soru: $x(t) = \frac{1}{1+t^2}$ $\quad X(f) = ?$

Reminder: $e^{-\alpha(t)} \longrightarrow \frac{2\alpha}{\alpha^2 + (2\pi f)^2}$

Duality $\quad x(t) \longrightarrow X(f)$
$X(t) \longrightarrow x(-f)$

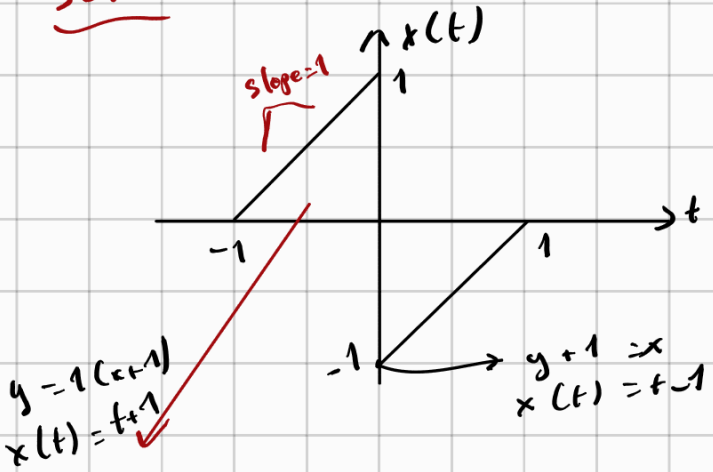
$\longrightarrow \frac{2\alpha}{4\pi^2}\left[\frac{1}{\frac{\alpha^2}{4\pi^2} + f^2}\right]$

$\frac{\alpha^2}{4\pi^2} = 1 \longrightarrow \alpha^2 = 4\pi^2 \Rightarrow \alpha = 2\pi$

$X(t) = \frac{2\cdot 2\pi}{4\pi^2}\left[\frac{1}{1+t^2}\right] = \frac{1}{\pi}\,\frac{1}{1+t^2}$

$x(-f) = e^{-\alpha|-f|} = e^{-\alpha|f|} = \pi \cdot e^{-\alpha|f|} = \pi \cdot e^{-2\pi|f|}$

Soru



$X(f) = ? = \int_{-\infty}^{\infty} x(t)\cdot e^{-j2\pi ft}\,dt$

$X(f) = \int_{-1}^{0} (t+1)\,e^{-j2\pi ft}\,dt + \int_{0}^{1} (t-1)\,e^{-j2\pi ft}\,dt$

sonuç $= \frac{j}{\pi f}\left[1 - \operatorname{sinc}(2f)\right]$

$= \underbrace{\int_{-1}^{0} t\,e^{-j2\pi ft}\,dt}_{I_1} + \underbrace{\int_{-1}^{0} e^{-j2\pi ft}\,dt}_{\frac{-1}{j2\pi f}\,e^{-j2\pi ft}\Big|_{-1}^{0}} + \underbrace{\int_{0}^{1} t\,e^{-j2\pi ft}\,dt}_{I_2} - \underbrace{\int_{0}^{1} e^{-j2\pi ft}\,dt}_{\frac{-1}{j2\pi f}\,e^{-j2\pi ft}\Big|_{0}^{1}}$

kısmi integrasyon

$$\int_a^b u\,dv = uv\Big|_a^b - \int_a^b v\,du$$

$$I_1 = \int_{-1}^{0} t e^{-j2\pi ft}\,dt =$$

$u = t$, $du = dt$

$dv = e^{-j2\pi ft}\,dt$, $v = \frac{1}{-j2\pi f} e^{-j2\pi ft}$

$$= \left(t \cdot \frac{-1}{j2\pi f} \right.$$

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